

Example: Finding the Volume Between Two Spheres

Benjamin Abshire[†]
Kent State University

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I Introduction

In order to find the volume between two spheres, we first must define how to write integrals in spherical coordinates. For this problem, we can do so as such:

$$\begin{aligned}\iiint_E f(x, y, z) \, dx \, dy \, dz &= \iiint_E f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi \\ &= \iiint_E f(\rho, \theta, \phi) \, dV\end{aligned}$$

Also, recall that r is equal to $x + y$, which is written in spherical coordinates as $\rho \sin \phi$.

II The Problem

Evaluate the integral

$$\iiint_E (x^2 + y^2) \, dV,$$

where E lies between spheres $x^2 + y^2 + z^2 = 9$ and $x^2 + y^2 + z^2 = 25$

Solution

First, observe that $(x^2 + y^2) = r^2 = \rho^2 \sin^2 \phi$ by our definition of r . Now that we have the integrand and bounds, we can write the rest of the integral using the rest of the definitions above:

$$\int_0^\pi \int_0^{2\pi} \int_3^5 (\rho^2 \sin^2 \phi) \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$$

Note that we choose 3 to 5 because those are the square roots of our constants in the sphere equations, and π and 2π for the spheres. Next, we can split up each integral and distribute ρ^2 and $\sin \phi$:

$$\int_0^\pi \sin^3 \phi \, d\phi \int_0^{2\pi} d\theta \int_3^5 \rho^4 \, d\rho$$

We get $\sin^3 \phi$ from multiplying $\sin^2 \phi$ by $\sin \phi$ and we get ρ^4 by multiplying ρ^2 by ρ^2 . There is no term with θ in the integrand, so we just integrate $d\theta$.

[†]Definitions and example problems from MATH 22005: Analytic Geometry and Calculus III instructed by Dr. Gang Yu

Next, we will evaluate that integral of $d\theta$ and move it out front:

$$\begin{aligned} & \int_0^\pi \sin^3 \phi \, d\phi \cdot \theta \Big|_0^{2\pi} \cdot \int_3^5 \rho^4 \, d\rho \\ &= 2\pi \int_0^\pi \sin^3 \phi \, d\phi \int_3^5 \rho^4 \, d\rho \end{aligned}$$

Now we can also evaluate the integral of ρ^4 and move it out front:

$$\begin{aligned} & 2\pi \int_0^\pi \sin^3 \phi \, d\phi \cdot \left(\frac{1}{5} \rho^5 \right) \Big|_3^5 \\ &= 2\pi \cdot \frac{2882}{5} \int_0^\pi \sin^3 \phi \, d\phi \\ &= \frac{5764\pi}{5} \int_0^\pi \sin^3 \phi \, d\phi \end{aligned}$$

From here we can use our knowledge of trig integrals from Calculus II and split off a $\sin \phi$ from $\sin^3 \phi$ and evaluate the integral:

$$\begin{aligned} & \frac{5764\pi}{5} \int_0^\pi \sin^2 \phi \sin \phi \, d\phi \\ &= \frac{5764\pi}{5} \int_0^\pi (1 - \cos^2 \phi) \sin \phi \, d\phi \end{aligned}$$

Let $u = \cos \phi$, $du = -\sin \phi \, d\phi \implies -du = \sin \phi \, d\phi$

$$\begin{aligned} & \text{So, } \frac{5764\pi}{5} \int_0^\pi -(1 - u^2) \, du \\ &= \frac{5764\pi}{5} \left(-u + \frac{1}{3} u^3 \right) \Big|_0^\pi \\ &= \frac{5764\pi}{5} \left(-\cos \phi + \frac{1}{3} \cos^3 \phi \right) \Big|_0^\pi \\ &= \frac{5764\pi}{5} \left(1 - \frac{1}{3} + 1 - \frac{1}{3} \right) \\ &= \frac{5764\pi}{5} \cdot \frac{4}{3} \\ &= \frac{23056\pi}{15} \end{aligned}$$

And we are done.